

(b) Values of d and λ are given in Table 2.1.

Table 2.1

$d/10^{21} \text{ km}$	$\frac{d}{c}/10^{15} \text{ s}$	λ/nm
0.48 ± 0.12		658.4
1.04 ± 0.12		661.2
1.45 ± 0.12		664.2
1.80 ± 0.12		665.7
2.85 ± 0.12		672.4
3.75 ± 0.12		678.2

The value of c is $3.00 \times 10^5 \text{ km s}^{-1}$.

Calculate and record values of $\frac{d}{c}/10^{15} \text{ s}$ in Table 2.1. Include the absolute uncertainties in $\frac{d}{c}$.
[2]

- (c) (i) Plot a graph of λ/nm against $\frac{d}{c}/10^{15} \text{ s}$. Include error bars for $\frac{d}{c}$. [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]
- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]

